

A quick overview of DevOps workshop

The entire process will includes the following steps:

1. Creating a repository on Github
2. Clone the repo on local system
3. Creating virtual server on AWS using EC2 service
4. Creating CI/CD pipelines on Github using Github actions

Steps to create a repo on Github:

1. Create account on github or sign in if already have
2. Click on new option to create a new repository
3. Give a repo name and add description(optional) and set the visibility public/private as per required
4. At last click on create repository

Steps to clone a repo on local system for development:

1. Copy the newly generated repo url from github
2. Open git bash and run the following command to clone
`git clone <repo_url>`
Ex: `git clone https://github.com/kumar-praveen/test.git`
A new directory will be created with the same name of repo
In this case it would be <test>
3. set the CWD to the newly created directory using command
`cd test`

Steps to push the code on github

1. After making changes in the check status using following command
`git status`
2. If any file is found changed the commit the change on github using following command
`git add <file_name>` → for a specific file
`git add .` → to add all the files
3. Now commit the changes using
`git commit -m 'message'`
- 3 Now at last push the code to github
`git push`

Steps to create instance of EC2 on AWS:

- 1 Create account on AWS or login if already have an account
2. Search EC2 in search bar and click on launch instance
3. Give a name to the machine
4. Select OS type (Ubuntu)
5. Select AMI - free tier
6. Select instance type - free tier(t3.micro)
7. Select key-pair or create a new one if you don't have and then select that
8. In network setting allow HTTPS, HTTP and SSH protocols
9. At last click on launch instance

Configuring CI/CD pipeline using Github actions

- 1 Go to settings on the same repo we have created
2. Go to Secrets and variables and click on actions
3. Create three new repository secret which includes
 - Name: EC2_HOST Value: <EC2_PUBLIC_IP_V4>
 - Name: EC2_SSH_KEY Value: <Content_of_pem_file>
 - Name: EC2_USER Value: <Username>

Now, whatever the changes will be made on local development will be deployed automatically